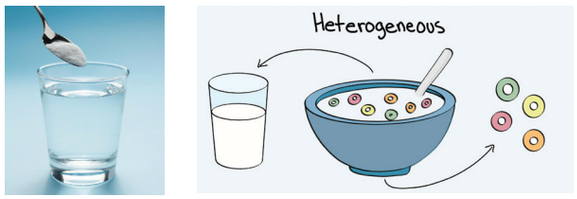
It has a uniform composition and every part of the mixture has the same properties. Tap water, milk, air, ice cream, sugar syrup, ink, steel, bronze and salt solution (Figure 10.4a) are homogeneous mixtures.

A mixture in which the components can be seen separately is called a heterogeneous mixture. It does not have a uniform composition and properties. Soil, a mixture of iodine and common salt, a mixture of sugar and sand, a mixture of oil and water, a mixture of sulphur and iron filings and a mixture of milk and cereals (Figure 10.4b) are heterogeneous mixture.




(a) (b)

**Figure 10.4** (a) Homogeneous and (b)Heterogeneous mixture

## 10.3 Separation of mixtures

Many mixtures contain useful substances mixed with unwanted material. In order to obtain these useful substances, chemists often have to separate them from the impurities. The choice of a particular method to separate components of a mixture will depend on the properties of the components of the mixture as well as their physical states (as shown in Table 10.3).

### 10.3.1 Sublimation

Certain solid substances when heated change directly from solid to gaseous state without attaining liquid state. The vapours when cooled give back the solid substance. This process is known as sublimation. Examples: Iodine, camphor, ammonium chloride etc.,

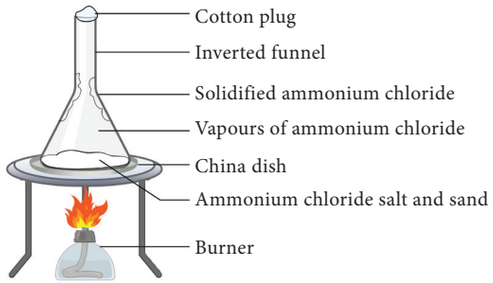



**Figure 10.5** Sublimation

The powdered mixture of Ammonium chloride and sand is taken in a china dish and covered with a perforated asbestos sheet. An inverted funnel is placed over the asbestos sheet as shown in Figure 10.5. The open end of the stem of the funnel is closed using cotton wool and the china dish is heated. The pure vapours of the volatile solid pass through the holes in the asbestos sheet and condense on the inner sides of the funnel. The non-volatile impurities remain in the china dish.

### More to Know

The air freshners are used in toilets. The solid slowly sublimes and releases the pleasant smell in the toilet over a certain period of time. Moth balls, made of naphthalene are used to drive away moths and some other insects. These also sublime over time. Camphor, is a substance used in Indian household. It sublimes to give a pleasant smell and is sometimes used as a freshner.

**Table 10.3** Methods of separating substances from mixtures

| Type of mixtues | Mixtures | Methods of separation |
|---|---|---|
| Heterogeneous | Solid and solid | Handpicking, sieving, winnowing, magnetic separation, **sublimation**. |
| | Insoluble solid and liquid | Sedimentation and decantation, loading, filtration, **centrifugation** |
| | Two immiscible liquids | Decantation, **solvent extraction** |
| Homogeneous | Soluble solid and liquid | Evaporation, **distillation**, crystallisation |
| | Two miscible liquids | **Fractional distillation** |
| | Solution of two or more solids in a liquid | **Chromatography** |



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